Digital System Design Project

Final Integrated System

UART-Controlled Digital System

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1 System Introduction

The final system is an integrated RTL digital system controlled through a UART interface.

The system receives commands from a master through the `RX_IN` UART interface. The received commands are processed by the system controller and can perform either Register File operations or ALU operations.

After the requested operation is completed, the system sends the corresponding result back to the master through the `TX_OUT` UART interface.

The system is divided into two main clock domains:

1. **REF_CLK domain**: responsible for system control, register file operations, ALU operations, and clock gating.
2. **UART_CLK domain**: responsible for UART transmission, UART reception, clock division, and pulse generation.

Since the system contains multiple clock domains, dedicated synchronization mechanisms are used to safely transfer reset, control, and data information between the domains.

2 System Specifications

The main system specifications are summarized in Table 1.

Table 1: System Specifications

Parameter	**Specification**
Reference Clock	100 MHz
UART Clock	3.6864 MHz
Clock Divider	Always enabled
UART Prescale	32 (default)
Data Width	8 bits
Register Address Width	4 bits
ALU Function Width	4 bits
UART Interface	RX/TX serial interface

2.1 System Block Diagram

Figure 1 shows the overall architecture of the integrated system.

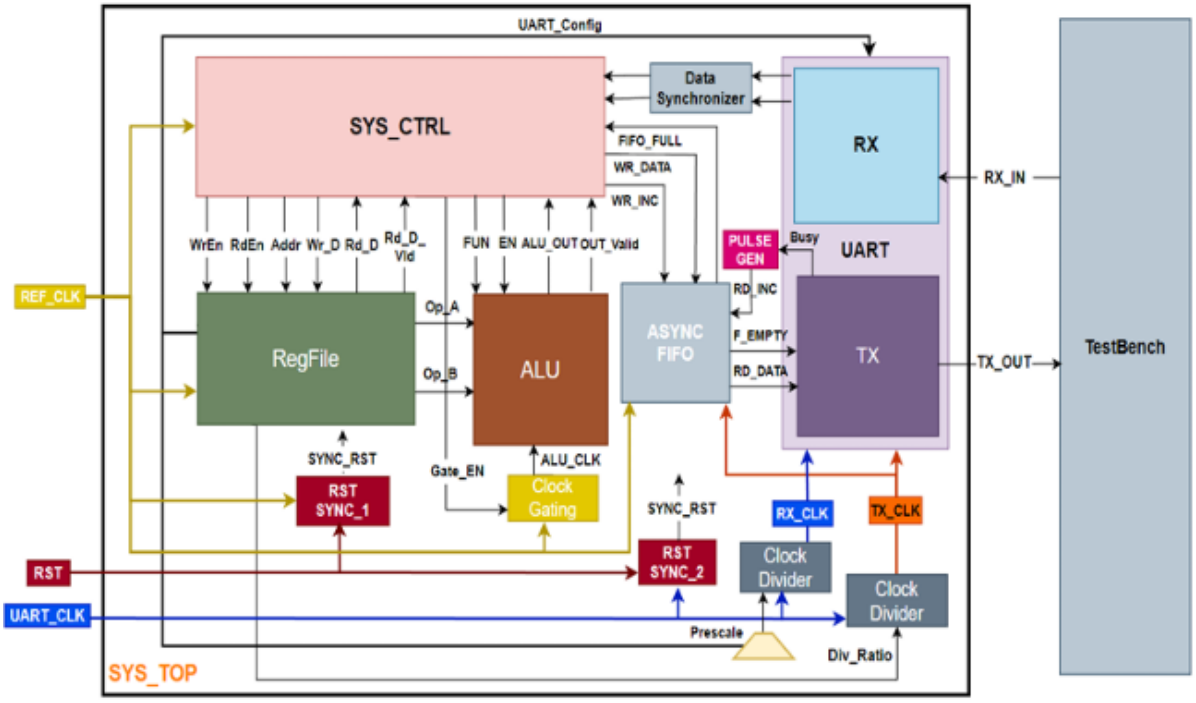


Figure 1: Overall architecture of the integrated UART-controlled digital system.

3 System Operation

The operational flow of the integrated system progresses through four main phases:

1. **Initialization & Setup:** Upon asserting the asynchronous active-low reset, all control state machines, counters, and pointers are forced into default states. The host then configures essential parameters—such as UART prescaling and clock division ratios—by writing to the reserved configuration registers.

2. **Command Ingestion & CDC:** The host transmits an 11-bit frame over the `RX_IN` serial interface. The `UART_RX` module samples the line and reconstructs the parallel byte. The `DATA_SYNC` block then safely transfers this data into the 100 MHz `REF_CLK` domain without metastability risks.

3. **Controller Decoding & Execution:** The `SYS_CTRL` state machine decodes the command opcode. For Register File access, it drives the target address and read/write enables. For ALU requests, it enables the clock gate (`CLK_GATE`), routes operands A and B, and triggers the ALU. The active unit computes the result and asserts `OUT_valid`.

4. **Response Transmission:** Output bytes are written into the `ASYNC_FIFO` to safely cross back into the `UART_CLK` domain. Finally, the `UART_TX` module pulls the data from the FIFO, serializes it into standard frames, and transmits the response back over `TX_OUT`.

4 Functional Blocks

4.1 ALU

The ALU (`ALU_TOP`) consists of four functional units: arithmetic, logic, comparison, and shift. A 2-to-4 decoder (`DECODER_2X4`), gated by the ALU enable from `SYS_CTRL`, activates one unit at a time. The 4-bit function code is split into two fields: `ALU_FUN[3:2]` selects the unit and `ALU_FUN[1:0]` selects the operation inside it. The operands A and B are 8 bits wide and come from `REG0` and `REG1` of the Register File.

- **Arithmetic:** signed addition, subtraction, multiplication, and division, producing a 16-bit signed result.
- **Logical:** AND, OR, NAND, and NOR, producing an 8-bit result.
- **Comparison:** compares A and B as unsigned values and returns a code.
- **Shift:** logical right and left shift of A or B by one bit, producing an 8-bit result. A zero enters the MSB on right shifts, and left shifts discard the MSB.

Table 2: ALU Function Codes

Code	Unit	Operation	Result width
`4'h0`	Arithmetic	$A + B$	16
`4'h1`	Arithmetic	$A - B$	16
`4'h2`	Arithmetic	$A \times B$	16
`4'h3`	Arithmetic	A/B	16
`4'h4`	Logic	$A \,\&\, B$ (AND)	8
`4'h5`	Logic	$A \mid B$ (OR)	8
`4'h6`	Logic	$\sim (A \,\&\, B)$ (NAND)	8
`4'h7`	Logic	$\sim (A \mid B)$ (NOR)	8
`4'h8`	Comparison	No operation (result 0)	8
`4'h9`	Comparison	$A = B \rightarrow 1$, else 0	8
`4'hA`	Comparison	$A > B \rightarrow 2$, else 0	8
`4'hB`	Comparison	$A < B \rightarrow 3$, else 0	8
`4'hC`	Shift	$A \gg 1$	8
`4'hD`	Shift	$A \ll 1$	8
`4'hE`	Shift	$B \gg 1$	8
`4'hF`	Shift	$B \ll 1$	8

Each unit registers its result on the (gated) ALU clock, so the result appears one clock cycle after the enable is asserted. The ALU then generates `OUT_valid`, a single-cycle pulse, to tell `SYS_CTRL` when to capture the result. Division by zero is the only illegal operation, and it produces no `OUT_valid` pulse. The 8-bit results are zero-extended to 16 bits when returned over UART.

4.2 Clock Gating

The clock-gating block (CLK_GATE) supplies the ALU clock. It takes the free-running REF_CLK and the enable ALU_CLK_EN generated by SYS_CTRL, and produces GATED_CLK, which clocks only the ALU. When the enable is low, the ALU clock is stopped, which removes unnecessary switching activity while the ALU is idle.

The block uses the standard latch-based structure:

- **Enable latch:** a level-sensitive latch that is transparent while REF_CLK is low and holds its value while REF_CLK is high.
- **AND gate:** combines REF_CLK with the latched enable to form GATED_CLK.

Because the latch only changes while the clock is low, a change on ALU_CLK_EN can never occur while REF_CLK is high. This guarantees a glitch-free gated clock with no truncated pulses, even though ALU_CLK_EN is generated combinationally by the SYS_CTRL state machine.
SYS_CTRL keeps ALU_CLK_EN high from the moment it starts waiting for the ALU function frame until the ALU result has been captured, so the clock is already running when the ALU enable pulse arrives and stays on until the result is latched. In all other states, including Register File commands, the ALU clock is stopped.

Two implementations of the block are provided:

1. **Behavioral model (simulation):** the latch and AND gate are written directly in Verilog, as described above.
2. **ASIC implementation:** the behavioral latch is replaced by the standard-cell integrated clock-gating cell TLATNCAX4M from the library (ports E, CK, ECK). This avoids the latch warning raised by lint on the behavioral version and ensures the gating structure is the one characterized by the library.

4.3 UART Receiver and Transmitter

Operating in the UART_CLK domain, the UART module handles bidirectional serial-parallel communication:

- **Receiver:** Converts incoming serial data on RX_IN into parallel output, generating a data-valid flag alongside parity and stop-bit error indicators.
- **Transmitter:** Packs parallel data into a standard serial frame—comprising a start bit, 8 data bits, an optional parity bit, and a stop bit—and outputs it to TX_OUT.

4.4 Pulse Generator

The pulse generator converts a level-type control signal into a single-cycle pulse used for FIFO read control.

4.5 Data Synchronizer

The Data Synchronizer (DATA_SYNC) enables safe, reliable transfer of multi-bit data vectors (unsync_bus) across asynchronous clock domain boundaries without data corruption or sampling hazards. The synchronization architecture functions in three sequential stages:

1. **Control-Path Metastability Settling:** The asynchronous enable input (bus_en) is fed into a multi-stage shift register (syncronizer) clocked by the destination domain clock (CLK). Parameterized by NUM_STAGES (defaulting to 8 stages for ultra-low failure probability), this shift register allows any potential metastable state induced at the boundary to settle into a stable logic level before reaching the output stage (syncronizer[0]).

2. **Edge Detection & Pulse Generation:** The bit output from the final synchronizer stage (syncronizer[0]) is registered into an internal flip-flop (pulse_gen) on every clock cycle to preserve its state from T_{n-1}. A combinational rising-edge detector computes en_pulse (syncronizer[0] $\wedge \neg$pulse_gen), which evaluates to high for exactly one destination clock cycle immediately following a $0 \rightarrow 1$ transition of the synchronized enable signal.

3. **Multiplexer-Gated Data Bus Latching:** The single-cycle en_pulse acts as a load-enable for the output register sync_bus. When en_pulse is asserted, the stable values on unsync_bus are latched into sync_bus. In all subsequent cycles where en_pulse is low, sync_bus holds its current value, ensuring the destination domain reads a fully coherent, non-glitching multi-bit data vector.

4.6 Asynchronous FIFO

The Asynchronous FIFO (ASYNC_FIFO) provides safe, full-duplex data transfer between independent, unsynchronized clock domains—specifically buffering data between the system domain (write_clk) and the UART domain (read_clk).
It utilizes a dual-port memory architecture coupled with Gray-coded pointer synchronization to guarantee glitch-free status flag generation and prevent data corruption during simultaneous read and write operations.

The internal operation is divided into four cohesive functional blocks:

- **Write Domain Controller (SOURCE_DOMAIN):** Operating on write_clk, this block manages memory allocation in the write domain

- **Read Domain Controller (DESTINATION_DOMAIN):** Operating on read_clk, this block manages memory retrieval in the read domain

- **Pointer Cross-Domain Synchronizers (GRAY_SYNC):** Two independent synchronizer instances (wr_2_rd and rd_2_wr) pass the Gray-coded pointers across the clock boundary. Parameterized by SYNC_STAGES (2-stage flip-flop chains), wr_2_rd synchronizes gray_write_ptr into read_clk, while rd_2_wr synchronizes gray_read_ptr into write_clk.

- **Dual-Port Storage Array (FIFO_BUFFER):** An internal static memory array (mem_array) parameterized by DATA_WIDTH and DEPTH.
  - Writes occur synchronously on the rising edge of write_clk at address write_addr, provided write_enable is active and fifo_full is low.
  - Reads are combinational; the data at location read_addr is driven directly onto rd_data, enabling immediate data availability without extra latency.

4.7 Clock Divider

The clock-divider module (CLK_DIV) operates in the UART clock domain and generates a lower-frequency clock (div_clk) from the reference clock (ref_clk) based on an 8-bit division ratio (div_ratio) programmed through the Register File.

The module features the following operational characteristics:

- **Enable and Bypass Mechanism:** The division logic is enabled via the active-high clk_en signal. When clk_en is deasserted, or when div_ratio is set to 0 or 1, the module bypasses the division counter and directly passes ref_clk to div_clk.
- **Even Division Ratios:** When div_ratio is even, an internal counter (counter_even) toggles the output clock every $\frac{\texttt{div_ratio}}{2}$ cycles of ref_clk, producing a clock with a 50% duty cycle.
- **Odd Division Ratios:** For odd division ratios, an internal state signal (odd_phase) alternates between a short phase of $\frac{\texttt{div_ratio}-1}{2}$ clock cycles and a long phase of $\frac{\texttt{div_ratio}-1}{2}+1$ clock cycles to achieve the correct average period.
- **Asynchronous Reset:** An active-low asynchronous reset (RST) resets all internal counters (counter_even, counter_odd), phase tracking (odd_phase), and sets the default output state of divided_clk_gen to high.

4.8 Register File

The Register File stores configuration values, ALU operands, and general-purpose register data.
The register address space is divided into reserved addresses and normal register addresses.

Table 3: Reserved Register Addresses

Address	Purpose	Description
0x0	ALU Operand A	ALU operand A
0x1	ALU Operand B	ALU operand B
0x2	UART Configuration	Parity and prescale configuration
0x3	Clock Divider	Clock division ratio

The UART configuration register contains:

- `REG2[0]`: Parity Enable
- `REG2[1]`: Parity Type
- `REG2[7:2]`: Prescale

The clock divider configuration is stored in:

$$\texttt{REG3[7:0]} = \text{Division Ratio}$$

Normal Register File operations use addresses from:

$$0x4 \rightarrow 0xF$$

4.9 SYS_CTRL

`SYS_CTRL` is the central controller of the system. It is a finite state machine (FSM) in the `REF_CLK` domain that receives each byte from the Data Synchronizer (`RX_P_DATA` with its valid flag `RX_D_VLD`), decodes the first frame of a command, and then drives the Register File, the ALU, the clock gate, and the asynchronous FIFO accordingly.

Its main responsibilities are:

- **Command decoding:** latches the first frame of every command and decodes it as `0xAA`, `0xBB`, `0xCC`, or `0xDD`. Any other opcode is ignored and the FSM stays in `IDLE`.
- **Register File control:** generates the address, write data, and the write/read enables for Register File commands. For `0xCC`, it also stores the two operand frames into `REG0` and `REG1`.
- **ALU control:** drives the function code and a single-cycle enable to the ALU, and captures the 16-bit result when the ALU asserts `OUT_valid`.
- **Clock-gating control:** enables the ALU clock only from the moment the function frame is awaited until the result has been captured.
- **Response generation:** writes the response bytes into the asynchronous FIFO: one byte for a register read, and two bytes (low byte first) for an ALU result. The UART transmitter reads them out on the `UART_CLK` side.

Table 4: SYS_CTRL FSM States

State	Function
IDLE	Waits for a valid frame and decodes it as a command opcode.
RF_ADDR	Latches the register address (4 LSBs of the frame).
RF_WR_DATA	Drives the address and, when the data frame arrives, asserts WrEn for one cycle.
RF_RD_WAIT	Asserts RdEn until RdData_valid, then pushes the read data into the FIFO.
OPERAND_A	Writes the received frame to REG0 (address 0x0).
OPERAND_B	Writes the received frame to REG1 (address 0x1).
ALU_FUN_WAIT	Enables the ALU clock and waits for the function frame. When it arrives, drives ALU_FUNC and pulses ALU_EN for one cycle.
ALU_EXEC	Holds the ALU clock and function code, and latches the 16-bit result. Moves to the FIFO states if OUT_valid is asserted, otherwise returns to IDLE.
FIFO_WRITE_LOW	Writes the low byte of the result into the FIFO when it is not full.
FIFO_WRITE_HIGH	Writes the high byte into the FIFO when it is not full, then returns to IDLE.

Table 5: State Sequence of Each Command

Command	State sequence
0xAA (RF write)	IDLE → RF_ADDR → RF_WR_DATA → IDLE
0xBB (RF read)	IDLE → RF_ADDR → RF_RD_WAIT → IDLE
0xCC (ALU with operands)	IDLE → OPERAND_A → OPERAND_B → ALU_FUN_WAIT → ALU_EXEC → FIFO_WRITE_LOW → FIFO_WRITE_HIGH → IDLE
0xDD (ALU without operands)	IDLE → ALU_FUN_WAIT → ALU_EXEC → FIFO_WRITE_LOW → FIFO_WRITE_HIGH → IDLE

The ALU enable is asserted for exactly one cycle, in the cycle the function frame arrives. The ALU registers its result on the next clock edge, so in ALU_EXEC the result is valid and OUT_valid is high. If OUT_valid is not asserted (division by zero), the FSM returns to IDLE and no response is sent.

5 UART Command Protocol

The system supports four command types.

5.1 Register File Write

The Register File Write command consists of three UART frames:

Table 6: Register File Write Command

Frame	Content	Value
Frame 0	Command	`0xAA`
Frame 1	Register Address	Address
Frame 2	Write Data	Data

5.2 Register File Read

The Register File Read command consists of two UART frames:

Table 7: Register File Read Command

Frame	Content	Value
Frame 0	Command	`0xBB`
Frame 1	Register Address	Address

After the command is processed, the requested register data is returned through the UART TX interface.

5.3 ALU Operation with Operands

The ALU operation with operands consists of four UART frames:

Table 8: ALU Operation with Operands

Frame	Content	Value
Frame 0	Command	`0xCC`
Frame 1	Operand A	8-bit data
Frame 2	Operand B	8-bit data
Frame 3	ALU Function	4-bit function code

5.4 ALU Operation without New Operands

The ALU operation without operands uses the operands already stored in the Register File.

It consists of two UART frames:

Table 9: ALU Operation without Operands

Frame	Content	Value
Frame 0	Command	0xDD
Frame 1	ALU Function	4-bit function code

6 Verification Methodology

The verification environment utilizes a self-checking Verilog testbench (SYS_TOP_tb) to validate the integrated SYS_TOP system from the external UART interface at the system level.

6.1 Verification Framework & Execution Flow

The self-checking testbench drives clock generation (REF_CLK at 100 MHz, UART_CLK at 3.6864 MHz, and uart_tx_clk_tb) and applies system reset while keeping RX_IN idle. It uses the send_frame task to package opcodes, addresses, and data into 11-bit UART frames (Start, 8 Data, Parity, Stop) driven into the DUT. For response capture, the receive_frame task detects the TX_OUT start bit and samples incoming serial data at exact bit centers. Finally, an automated checking mechanism compares the reconstructed output bytes against expected values using exact 4-state logic (===), logging results to Pass/Fail counters.

7 Main Command Tests

This section shows each of the four main commands at frame level. One waveform is given per frame, covering the start bit, 8 data bits, parity bit, and stop bit.

7.1 Register Write (`CMD_REG_WRITE`)

Objective: Verify that the opcode `8'hAA` followed by an address and a data byte writes the data into the Register File. The command takes 3 RX frames and returns no response.

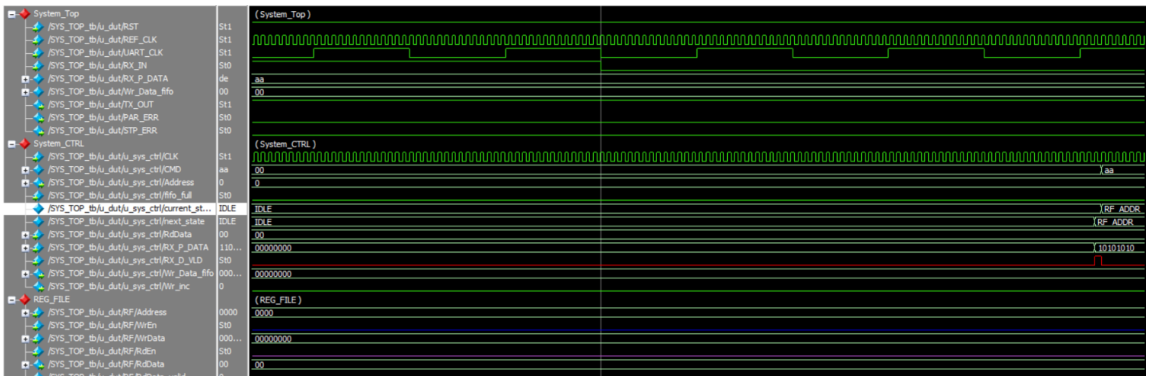


Figure 2: Register Write – Frame 1: opcode `0xAA`.

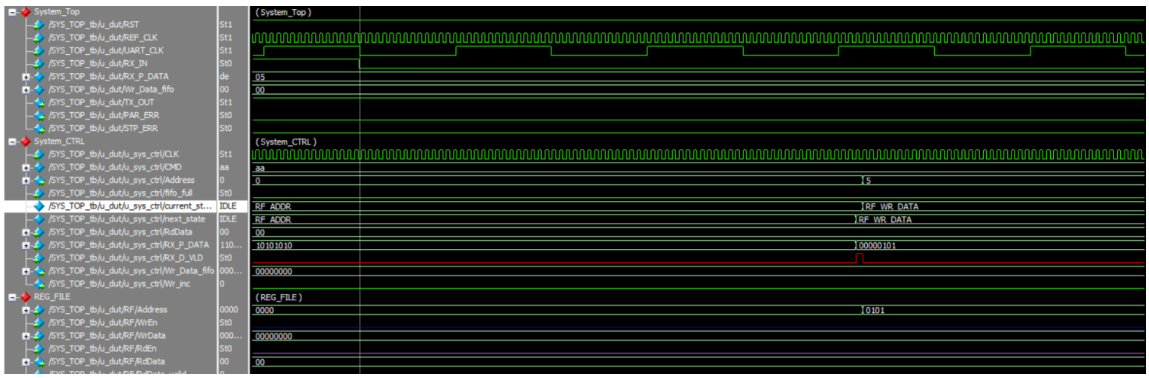


Figure 3: Register Write – Frame 2: address `0x05`.

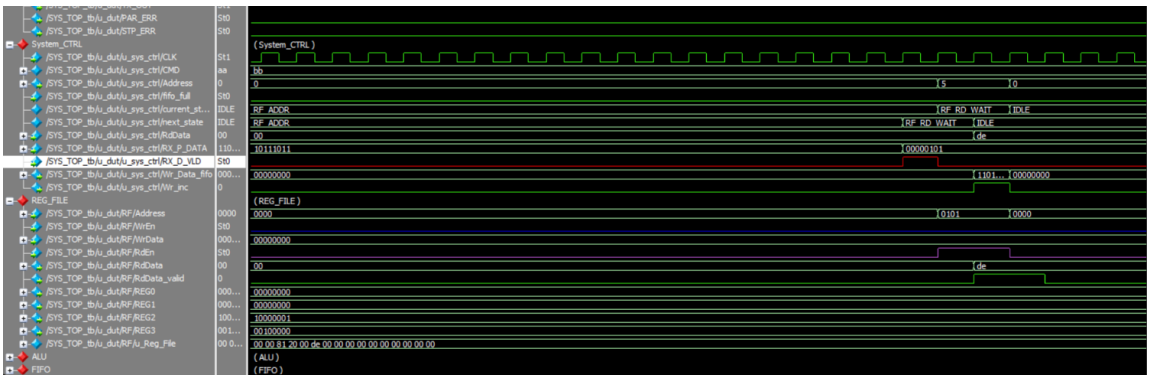


Figure 4: Register Write – Frame 3: data `0xDE`.

7.2 Register Read (CMD_REG_READ)

Objective: Verify that the opcode `8'hBB` followed by an address returns the stored value on `TX_OUT`. The command takes 2 RX frames and 1 TX frame. Reading address `0x05` returns `0xDE`, which confirms the previous write.

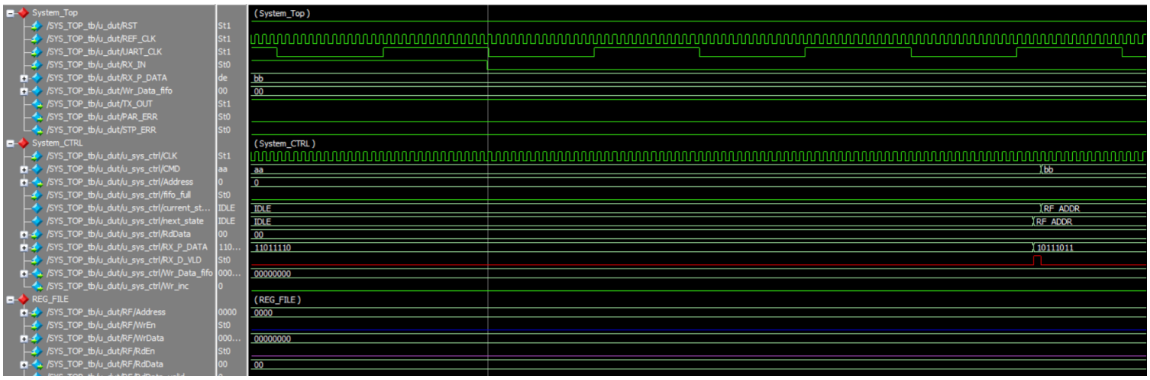


Figure 5: Register Read – Frame 1: opcode `0xBB`.

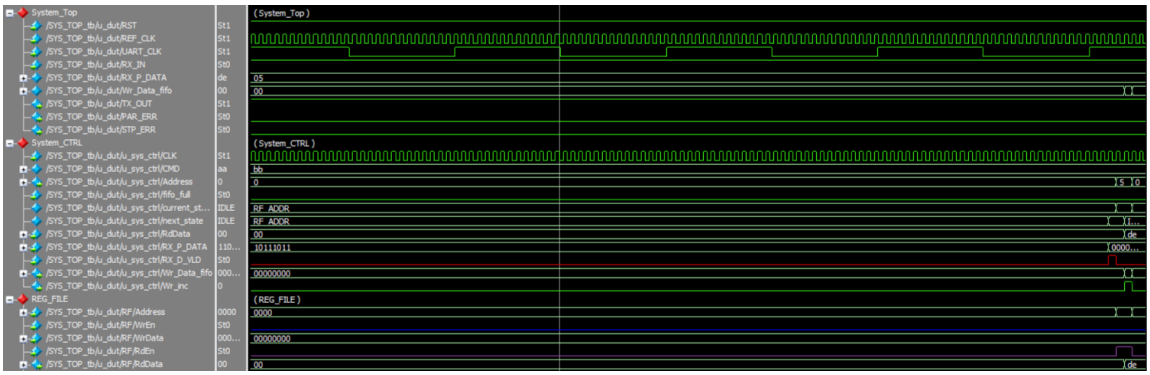


Figure 6: Register Read – Frame 2: address `0x05`.

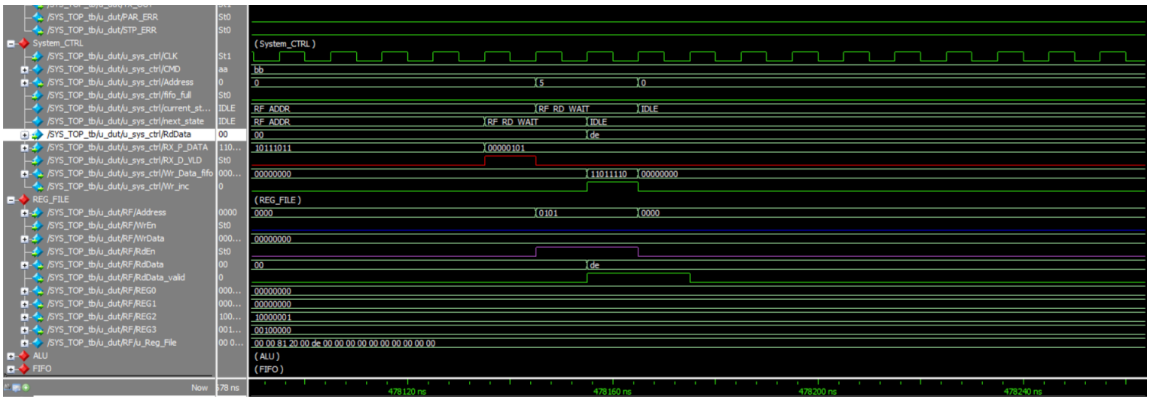


Figure 7: Register Read – Response:`0xDE`.

7.3 ALU Operation with Operands (`CMD_ALU_OPERANDS`)

Objective: Verify that the opcode `8'hCC` captures operands A and B and the function code, executes the operation, and returns the 16-bit result (Low Byte first, High Byte second). The command takes 4 RX frames and 2 TX frames. The example below is a signed multiplication of two negative operands, $(-6) \times (-7) = 42$, where $A =$ `0xFA` and $B =$ `0xF9` give `0x002A`.

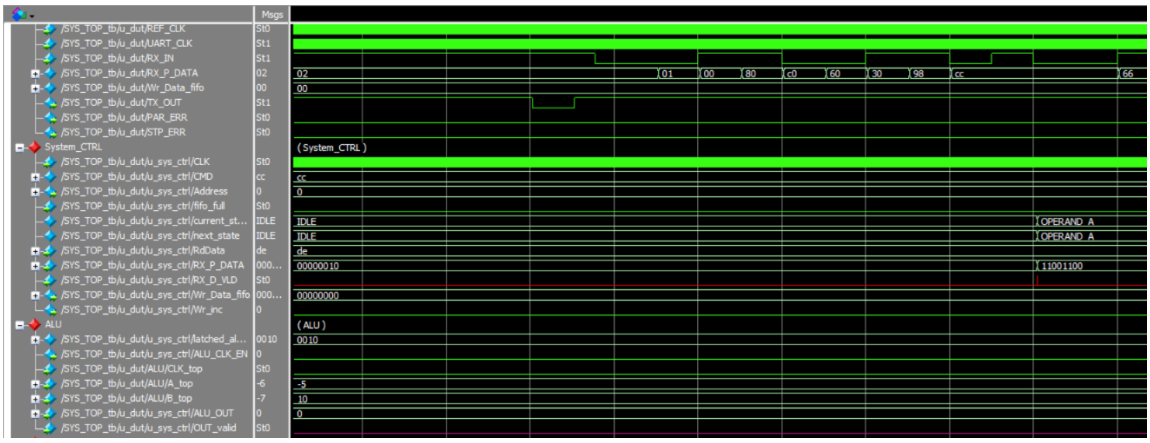


Figure 8: ALU with Operands – Frame 1: opcode `0xCC`.

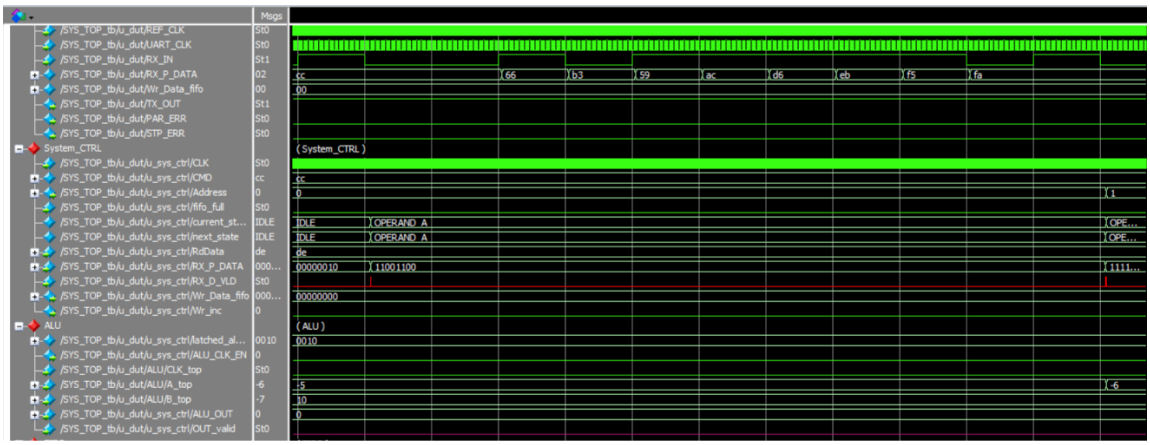


Figure 9: ALU with Operands – Frame 2: operand $A =$ `0xFA` (-6).

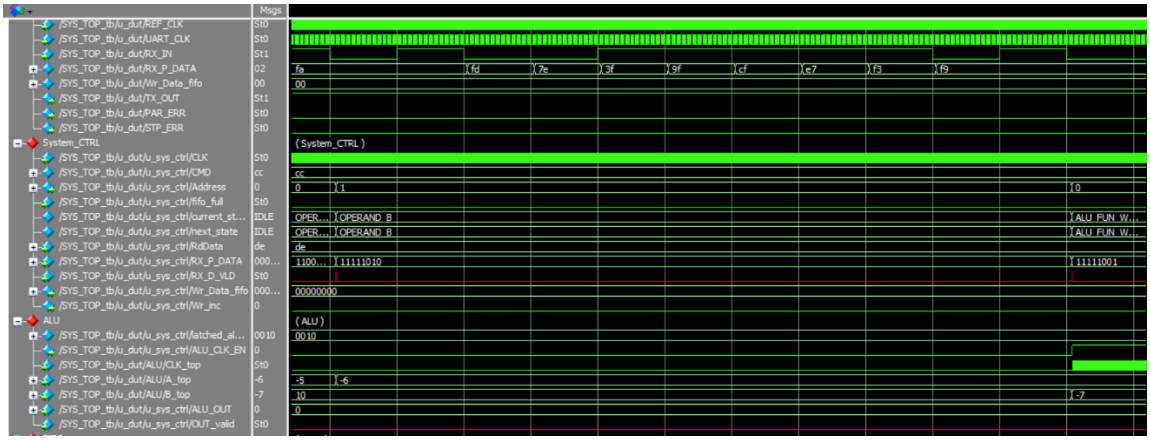


Figure 10: ALU with Operands – Frame 3: operand B = `0xF9` (-7).

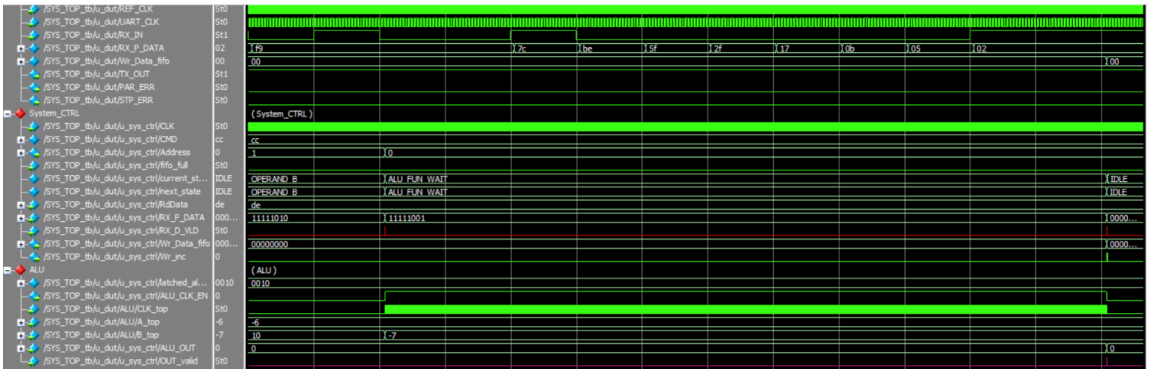


Figure 11: ALU with Operands – Frame 4: function code `4'h2` (MUL).

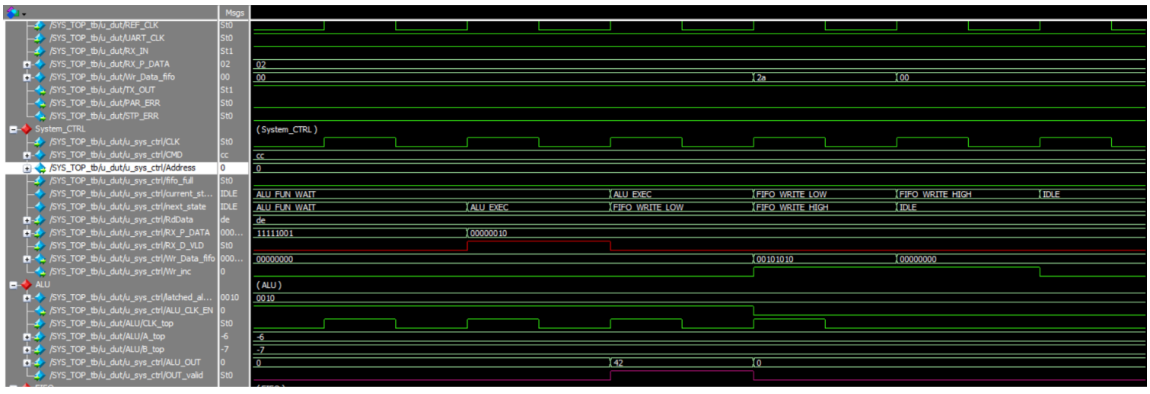


Figure 12: ALU with Operands – Response (Low Byte - High Byte): `0x2A`.

7.4 ALU Operation without Operands (`CMD_ALU_NOP`)

Objective: Verify that the opcode `8'hDD` followed by a function code executes the operation on the operands already stored in the Register File and returns the 16-bit result. The command takes 2 RX frames and 2 TX frames.

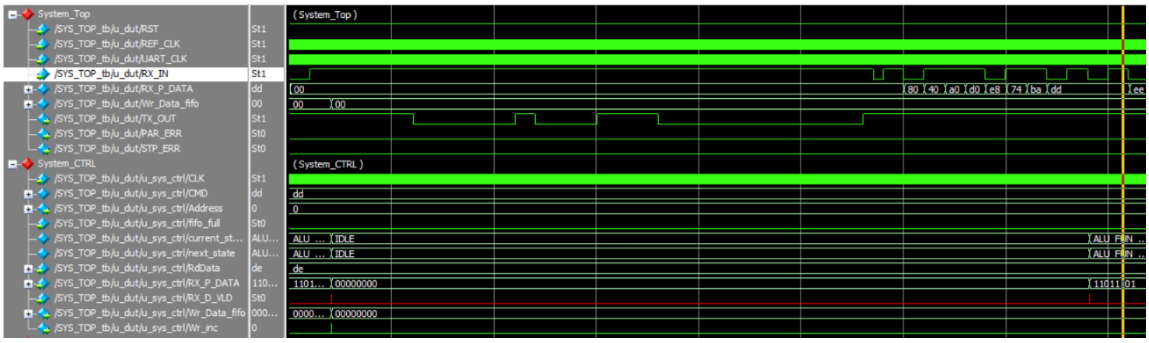


Figure 13: ALU without Operands – Frame 1: opcode `0xDD`.

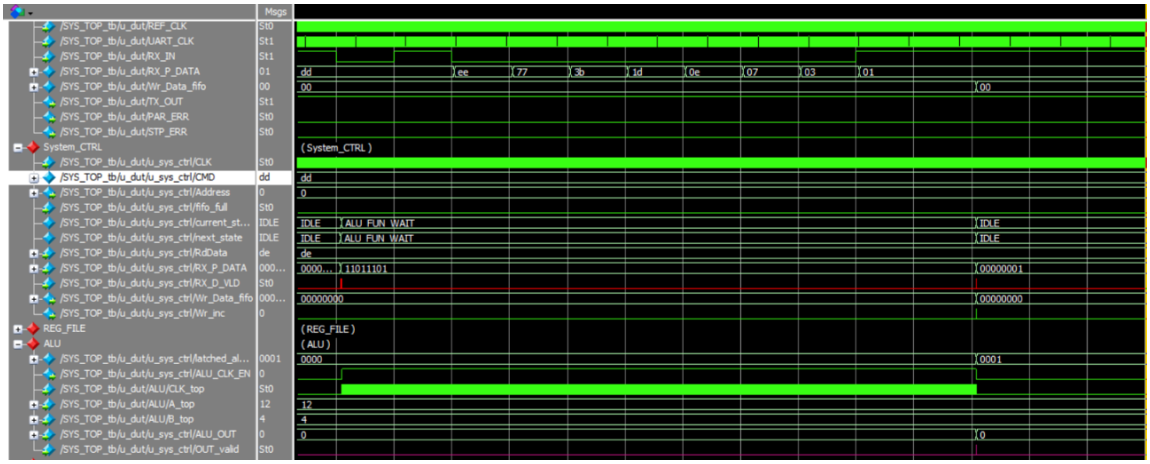


Figure 14: ALU without Operands – Frame 2: function code.

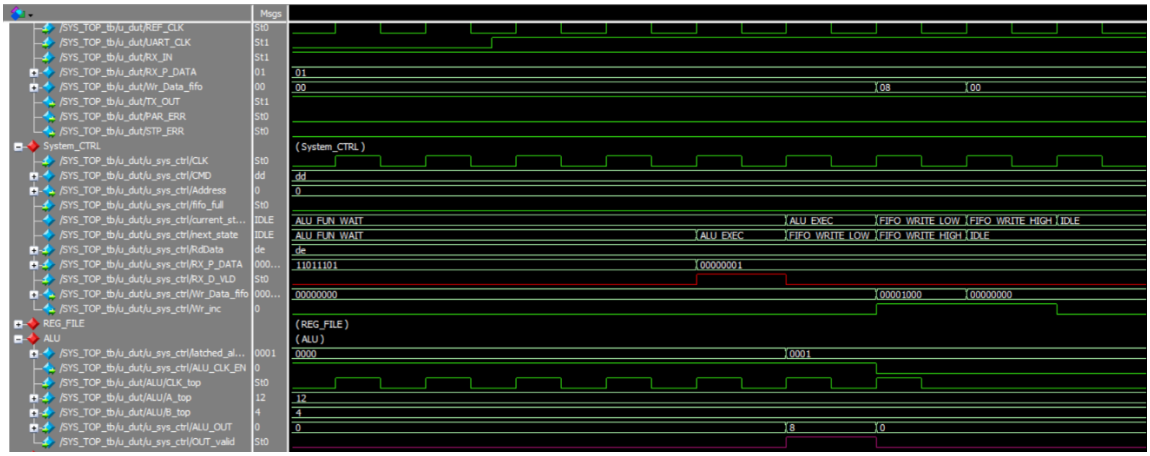


Figure 15: ALU without Operands – Response (Low Byte - High Byte).

8 Remaining Tests

Besides the main command captures, the testbench runs the remaining ALU vectors through `CMD_ALU_OPERANDS` (`8'hCC`) and `CMD_ALU_NO_OPERANDS` (`8'hDD`). Each check sends the opcode (and operands where applicable), then compares the 16-bit result against the expected value. These vectors are not shown as waveforms; The Register File write and read-back (Group 1) is the same transaction shown in the first two main command tests.

8.1 Group 2 – Logic Unit

Five vectors cover AND, OR, NAND, and NOR with negative and positive operands. Results are zero-extended to 16 bits. The last AND vector checks that the sign bit of A does not leak into the upper byte.

Table 10: Logic Unit Test Vectors

#	Operation	A	B	Func	Expected / Got
1	AND	`0xF0` (-16)	`0xAA` (-86)	`4'h4`	`0x00A0` (160)
2	OR	`0xF0` (-16)	`0x0F` (15)	`4'h5`	`0x00FF` (255)
3	NAND	`0xF0` (-16)	`0xAA` (-86)	`4'h6`	`0x005F` (95)
4	NOR	`0xF0` (-16)	`0x0F` (15)	`4'h7`	`0x0000` (0)
5	AND	`0xF6` (-10)	`0x0F` (15)	`4'h4`	`0x0006` (6)

8.2 Group 3 – Comparison Unit

These vectors check the three compare outcomes (equal $\rightarrow$ 1, greater $\rightarrow$ 2, less $\rightarrow$ 3). They also cover two negative cases: the no-operation code (`4'h8`) and an equality compare of unequal operands, both of which must return `0x0000`.

Table 11: Comparison Unit Test Vectors

#	Operation	A	B	Func	Expected / Got
1	$A = B$ (equal)	`0x0A` (10)	`0x0A` (10)	`4'h9`	`0x0001` (1)
2	$A > B$ (greater)	`0x14` (20)	`0x0A` (10)	`4'hA`	`0x0002` (2)
3	$A < B$ (less)	`0x05` (5)	`0x0F` (15)	`4'hB`	`0x0003` (3)
4	No operation	`0x14` (20)	`0x0A` (10)	`4'h8`	`0x0000` (0)
5	$A = B$ (not equal)	`0xFB` (-5)	`0x0A` (10)	`4'h9`	`0x0000` (0)

8.3 Group 4 – Shift Unit

The shifts cover $A \gg 1$, $A \ll 1$, $B \gg 1$, and $B \ll 1$. The right shifts are logical: `0x80` $\gg 1 =$ `0x0040` and `0xF6` $\gg 1 =$ `0x007B` show that a zero enters the MSB even for negative operands.

Table 12: Shift Unit Test Vectors

#	Operation	A	B	Func	Expected / Got
1	$A \gg 1$	`0x80` (-128)	`0x00` (0)	`4'hC`	`0x0040` (64)
2	$A \ll 1$	`0x0F` (15)	`0x00` (0)	`4'hD`	`0x001E` (30)
3	$B \gg 1$	`0x00` (0)	`0xAA` (-86)	`4'hE`	`0x0055` (85)
4	$B \ll 1$	`0x00` (0)	`0x55` (85)	`4'hF`	`0x00AA` (170)
5	$A \gg 1$	`0xF6` (-10)	`0x00` (0)	`4'hC`	`0x007B` (123)

8.4 Group 5 – Signed Arithmetic

Subtraction, addition, and multiplication are checked with mixed-sign operands. Results are sign-extended to 16 bits, and the $(-6) \times (-7)$ vector covers the negative×negative case.

Table 13: Signed Arithmetic Test Vectors

#	Operation	A	B	Func	Expected / Got
1	SUB	`0x05` (5)	`0x0A` (10)	`4'h1`	`0xFFFB` (-5)
2	ADD	`0xF6` (-10)	`0x19` (25)	`4'h0`	`0x000F` (15)
3	MUL	`0xFB` (-5)	`0x0A` (10)	`4'h2`	`0xFFCE` (-50)
4	MUL	`0xFA` (-6)	`0xF9` (-7)	`4'h2`	`0x002A` (42)

8.5 Group 6 – ALU No-Operands Tests

These vectors execute ALU operations using the existing values stored in the Register File (`REG0` = `12` and `REG1` = `4`). They verify that operations trigger properly using stored state without requiring inline operands.

Table 14: ALU No-Operands Test Vectors (`REG0`=12, `REG1`=4)

#	Operation	Func	Formula	Expected / Got
1	ADD	`4'h0`	$12 + 4$	`0x0010` (16)
2	SUB	`4'h1`	$12 - 4$	`0x0008` (8)
3	MUL	`4'h2`	12×4	`0x0030` (48)
4	DIV	`4'h3`	$12/4$	`0x0003` (3)
5	AND	`4'h4`	`0x0C` & `0x04`	`0x0004` (4)

9 Conclusion

In this project, a fully integrated, robust RTL digital system controlled via a serial UART interface was successfully designed, implemented, and verified. The architecture efficiently orchestrates operations across two distinct clock domains—the high-frequency system reference domain (`REF_CLK` at 100 MHz) and the lower-frequency serial communication domain (`UART_CLK` at 3.6864 MHz).

Key technical achievements and architectural highlights include:

- **Robust Clock Domain Crossing (CDC):** By employing a multi-stage Data Synchronizer with recirculating enable logic for control signals and an Asynchronous FIFO driven by Gray-coded pointers, the system guarantees glitch-free, metastability-resistant data transfers across independent clock domains.
- **Power Efficiency via Dynamic Clock Gating:** An integrated latch-based clock-gating cell (`CLK_GATE`) ensures that the ALU clock is dynamically enabled only during active computation cycles, effectively eliminating unnecessary dynamic switching power while the processing unit is idle.
- **Flexible and Scalable Interface:** The finite state machine within `SYS_CTRL` seamlessly decodes a multi-frame UART protocol to execute register configuration, direct memory access, and multi-mode 16-bit ALU operations (arithmetic, logical, comparison, and shifting).
- **Exhaustive Verification:** Using an automated, self-checking Verilog testbench, all core commands (`0xAA`, `0xBB`, `0xCC`, and `0xDD`) along with edge-case ALU test vectors were verified at bit-level precision, confirming 100% functional correctness against expected multi-byte responses.

Overall, the design meets all target specifications, proving to be a reliable and energy-efficient solution for UART-driven peripheral and processing control.